

Experiment (7)

Wide Area Network (WAN)

7.1 Objectives

To achieve Encapsulation between 2 Routers

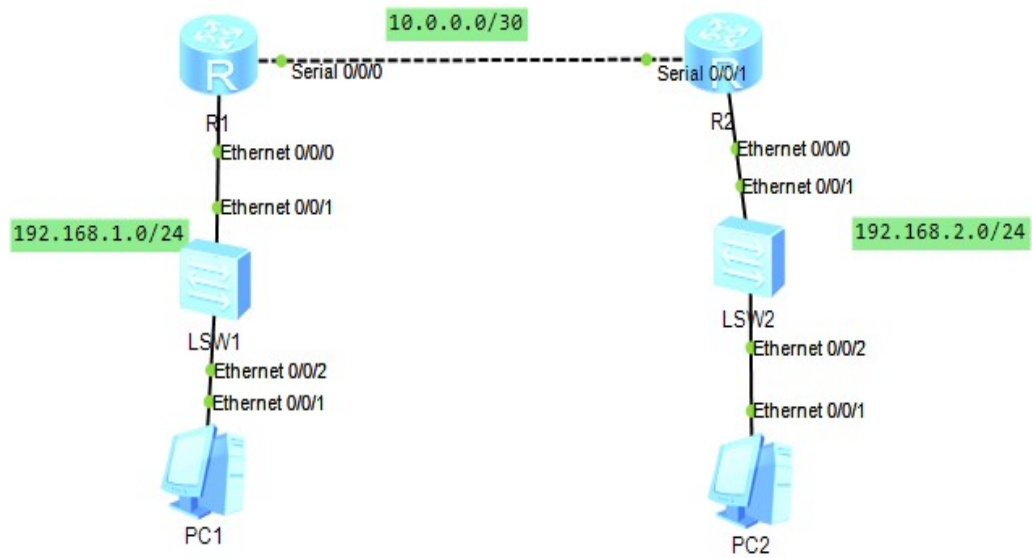
7.2 Equipment / Program Requirements:

ENSP

7.3 Experiment Setup:

Make topology as shown then make encapsulation then ping between 2 routers to check connections

7.4 Simulation Results:



R1

```
The device is running!
#####
#####
<Huawei>
Apr 17 2026 19:30:53-08:00 Huawei %%01PHY/1/PHY(1)[0]: Ethernet0/0/0: change
status to up
Apr 17 2026 19:30:53-08:00 Huawei %%01PHY/1/PHY(1)[1]: Serial0/0/0: change st
atus to up
Apr 17 2026 19:30:56-08:00 Huawei %%01IFNET/4/LINK_STATE(1)[2]:The line protocol
PPP on the interface Serial0/0/0 has entered the UP state.
<Huawei>u t l
Info: Current terminal logging is off.
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]int s0/0/0
[Huawei-Serial0/0/0]ip add
[Huawei-Serial0/0/0]ip address 10.0.0.1 30
[Huawei-Serial0/0/0]int eth0/0/0
[Huawei-Ethernet0/0/0]ip add
[Huawei-Ethernet0/0/0]ip address 192.168.1.1 24
[Huawei-Ethernet0/0/0]int s0/0/0
[Huawei-Serial0/0/0]lin
[Huawei-Serial0/0/0]link-protocol hdlc
Warning: The encapsulation protocol of the link will be changed.
Continue? [Y/N]:Y
[Huawei-Serial0/0/0]display interface serial 0/0/0
Serial0/0/0 current state : UP
Line protocol current state : UP
Last line protocol up time : 2026-04-17 19:34:30 UTC-08:00
Description:
Route Port,The Maximum Transmit Unit is 1500, Hold timer is 10(sec)
Internet Address is 10.0.0.1/30
Link layer protocol is nonstandard HDLC
Last physical up time : 2026-04-17 19:34:30 UTC-08:00
Last physical down time : 2026-04-17 19:34:29 UTC-08:00
Current system time: 2026-04-17 19:35:13-08:00Interface is V35
Last 300 seconds input rate 0 bytes/sec, 0 packets/sec
Last 300 seconds output rate 0 bytes/sec, 0 packets/sec
Input: 676 bytes, 56 Packets
Output: 702 bytes, 53 Packets
Input bandwidth utilization : 0%
Output bandwidth utilization : 0%
```



Type here to search



R1

```
[Huawei-Serial0/0/0]ip address 10.0.0.1 30
[Huawei-Serial0/0/0]int eth0/0/0
[Huawei-Ethernet0/0/0]ip add
[Huawei-Ethernet0/0/0]ip address 192.168.1.1 24
[Huawei-Ethernet0/0/0]int s0/0/0
[Huawei-Serial0/0/0]lin
[Huawei-Serial0/0/0]link-protocol hdlc
Warning: The encapsulation protocol of the link will be changed.
Continue? [Y/N]:Y
[Huawei-Serial0/0/0]display interface serial 0/0/0
Serial0/0/0 current state : UP
Line protocol current state : UP
Last line protocol up time : 2026-04-17 19:34:30 UTC-08:00
Description:
Route Port,The Maximum Transmit Unit is 1500, Hold timer is 10(sec)
Internet Address is 10.0.0.1/30
Link layer protocol is nonstandard HDLC
Last physical up time : 2026-04-17 19:34:30 UTC-08:00
Last physical down time : 2026-04-17 19:34:29 UTC-08:00
Current system time: 2026-04-17 19:35:13-08:00Interface is V35
  Last 300 seconds input rate 0 bytes/sec, 0 packets/sec
  Last 300 seconds output rate 0 bytes/sec, 0 packets/sec
  Input: 676 bytes, 56 Packets
  Output: 702 bytes, 53 Packets
  Input bandwidth utilization : 0%
  Output bandwidth utilization : 0%

[Huawei-Serial0/0/0]
[Huawei-Serial0/0/0]ping 10.0.0.2
  PING 10.0.0.2: 56 data bytes, press CTRL_C to break
    Reply from 10.0.0.2: bytes=56 Sequence=1 ttl=255 time=40 ms
    Reply from 10.0.0.2: bytes=56 Sequence=2 ttl=255 time=30 ms
    Reply from 10.0.0.2: bytes=56 Sequence=3 ttl=255 time=1 ms
    Reply from 10.0.0.2: bytes=56 Sequence=4 ttl=255 time=30 ms
    Reply from 10.0.0.2: bytes=56 Sequence=5 ttl=255 time=10 ms

  --- 10.0.0.2 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 1/22/40 ms

[Huawei-Serial0/0/0]ip route-static 192.168.2.0 24 10.0.0.2
[Huawei]
```



Type here to search



R2

```
Last physical up time : -
Last physical down time : 2026-04-17 19:30:50 UTC-08:00
Current system time: 2026-04-17 19:35:36-08:00Interface is V35
  Last 300 seconds input rate 0 bytes/sec, 0 packets/sec
  Last 300 seconds output rate 0 bytes/sec, 0 packets/sec
  Input: 0 bytes, 0 Packets
  Output: 0 bytes, 0 Packets
  Input bandwidth utilization : 0%
  Output bandwidth utilization : 0%

[Huawei-Serial0/0/1]display interface serial 0/0/1
Serial0/0/1 current state : UP
Line protocol current state : UP
Last line protocol up time : 2026-04-17 19:35:33 UTC-08:00
Description:
Route Port,The Maximum Transmit Unit is 1500, Hold timer is 10(sec)
Internet Address is 10.0.0.2/30
Link layer protocol is nonstandard HDLC
Last physical up time : 2026-04-17 19:35:33 UTC-08:00
Last physical down time : 2026-04-17 19:35:33 UTC-08:00
Current system time: 2026-04-17 19:36:29-08:00Interface is V35
  Last 300 seconds input rate 2 bytes/sec, 0 packets/sec
  Last 300 seconds output rate 2 bytes/sec, 0 packets/sec
  Input: 838 bytes, 59 Packets
  Output: 826 bytes, 66 Packets
  Input bandwidth utilization : 0.02%
  Output bandwidth utilization : 0.02%

[Huawei-Serial0/0/1]ping 10.0.0.2
PING 10.0.0.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.0.2: bytes=56 Sequence=1 ttl=255 time=40 ms
  Reply from 10.0.0.2: bytes=56 Sequence=2 ttl=255 time=1 ms
  Reply from 10.0.0.2: bytes=56 Sequence=3 ttl=255 time=20 ms
  Reply from 10.0.0.2: bytes=56 Sequence=4 ttl=255 time=30 ms
  Reply from 10.0.0.2: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 10.0.0.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/22/40 ms

[Huawei-Serial0/0/1]ip route-static 192.168.1.0 24 10.0.0.1
[Huawei]
```

 R2

```
Last physical up time : -
Last physical down time : 2026-04-17 19:30:50 UTC-08:00
Current system time: 2026-04-17 19:35:36-08:00Interface is V35
  Last 300 seconds input rate 0 bytes/sec, 0 packets/sec
  Last 300 seconds output rate 0 bytes/sec, 0 packets/sec
  Input: 0 bytes, 0 Packets
  Output: 0 bytes, 0 Packets
  Input bandwidth utilization : 0%
  Output bandwidth utilization : 0%

[Huawei-Serial0/0/1]display interface serial 0/0/1
Serial0/0/1 current state : UP
Line protocol current state : UP
Last line protocol up time : 2026-04-17 19:35:33 UTC-08:00
Description:
Route Port,The Maximum Transmit Unit is 1500, Hold timer is 10(sec)
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  Last 300 seconds input rate 2 bytes/sec, 0 packets/sec
  Last 300 seconds output rate 2 bytes/sec, 0 packets/sec
  Input: 838 bytes, 59 Packets
  Output: 826 bytes, 66 Packets
  Input bandwidth utilization : 0.02%
  Output bandwidth utilization : 0.02%

[Huawei-Serial0/0/1]ping 10.0.0.2
PING 10.0.0.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.0.2: bytes=56 Sequence=1 ttl=255 time=40 ms
  Reply from 10.0.0.2: bytes=56 Sequence=2 ttl=255 time=1 ms
  Reply from 10.0.0.2: bytes=56 Sequence=3 ttl=255 time=20 ms
  Reply from 10.0.0.2: bytes=56 Sequence=4 ttl=255 time=30 ms
  Reply from 10.0.0.2: bytes=56 Sequence=5 ttl=255 time=20 ms

--- 10.0.0.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/22/40 ms

[Huawei-Serial0/0/1]ip route-static 192.168.1.0 24 10.0.0.1
[Huawei]
```

R2

The device is running!

```
<Huawei>u t l
Info: Current terminal logging is off.
<Huawei>u t m
Info: Current terminal monitor is off.
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]int se0/0/1
[Huawei-Serial0/0/1]ip add
[Huawei-Serial0/0/1]ip address 10.0.0.2 30
[Huawei-Serial0/0/1]int eth0/0/0
[Huawei-Ethernet0/0/0]ip add
[Huawei-Ethernet0/0/0]ip address 192.168.2.1 24
[Huawei-Ethernet0/0/0]int se0/0/1
[Huawei-Serial0/0/1]li
[Huawei-Serial0/0/1]link-protocol hdlc
Warning: The encapsulation protocol of the link will be changed.
Continue? [Y/N]:Y
```

```
PC>ping 192.168.2.2

Ping 192.168.2.2: 32 data bytes, Press Ctrl_C to break
From 192.168.2.2: bytes=32 seq=1 ttl=126 time=156 ms
From 192.168.2.2: bytes=32 seq=2 ttl=126 time=125 ms
From 192.168.2.2: bytes=32 seq=3 ttl=126 time=109 ms
From 192.168.2.2: bytes=32 seq=4 ttl=126 time=125 ms
From 192.168.2.2: bytes=32 seq=5 ttl=126 time=140 ms

--- 192.168.2.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 109/131/156 ms

PC>|
```

7.5 Conclusions:

Encapsulation Is important for 2 routers to move data between 2 routers